

XGBoost for Classification

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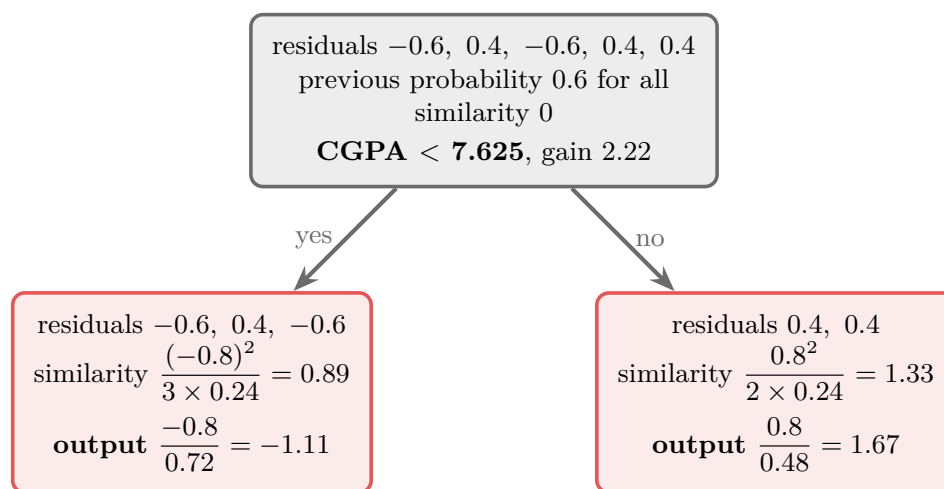
Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Binning and binarization ([Note 32](#)); Regularisation ([Note 63](#)); Gradient boosting ([Note 122](#)); Missing values ([Note 123](#)).

1. Overview

Key point: XGBoost classifies like gradient boosting for classification: start from the log-odds, turn it into a probability, fit trees to the residuals $y - p$, and add their outputs in log-odds. Its trees use the similarity score and gain from regression, with $\sum p(1 - p)$ in place of the number of residuals.



$\lambda = 0$; outputs are in log-odds

Figure 1: The first XGBoost classification tree on five students: residuals, similarity scores and leaf outputs

This Note joins two earlier ones. The flow (log-odds, probabilities, residuals) comes from the [gradient boosting classification Note](#). The tree (similarity score, gain, output value) comes from the [XGBoost regression Note](#).

The one real change is the denominator of the formulas, shown in Figure 1. Where the formulas come from is derived in the [XGBoost maths Note](#) (Chen and Guestrin 2016, §2.2).

2. The data: CGPA and placement

Key point: Five students, one feature (CGPA) and a class target: placed (1) or not (0). Three were placed, two were not.

Student	CGPA	Placed
1	5.70	0
2	6.25	1
3	7.10	0
4	8.15	1
5	9.60	1

Each student is one **observation** (one record, a row of the table). CGPA is the only **feature** (an input variable, one column of the table), and “Placed” is the **target** (the output we predict).

Given a new student’s CGPA, the model must say whether they will be placed. The table is already sorted by CGPA, which we will need for the splits.

3. Stage 1: the log-odds and its probability

Key point: The base model predicts the log-odds of placement, $\ln(3/2) = 0.405$, for everyone. As a probability that is 0.6.

As in the [gradient boosting classification Note](#), sections 4 and 5, stage 1 predicts the log-odds of class 1, $\ln(p/(1 - p))$, and the sigmoid turns it back into a probability. Here 3 of the 5 students were placed, so $p = 3/5$:

$$f_0 = \ln \frac{3/5}{2/5} = \ln 1.5 = 0.405, \quad p = \frac{e^{0.405}}{1 + e^{0.405}} = \frac{1.5}{2.5} = 0.6$$

So stage 1 says “probability 0.6” for every student, whatever the CGPA. With a threshold of 0.5 it predicts “placed” for all five, which is wrong for students 1 and 3.

4. Residuals in probability

Key point: Residual = actual class – predicted probability: -0.6 for the two students not placed, $+0.4$ for the three placed.

Student	CGPA	Placed	Probability 1	Residual 1
1	5.70	0	0.6	-0.6
2	6.25	1	0.6	0.4

Student	CGPA	Placed	Probability 1	Residual 1
3	7.10	0	0.6	-0.6
4	8.15	1	0.6	0.4
5	9.60	1	0.6	0.4

Residuals are always computed from probabilities, never from log-odds. The next model, an XGBoost tree, takes CGPA as its feature and these residuals as its target.

5. The similarity score for classification

Key point: Similarity = (sum of residuals)² divided by $(\sum p(1-p) + \lambda)$, where p is each observation's previous probability. The root scores 0.

The numerator is the same as in regression. The denominator replaces “number of residuals” with the sum of $p(1-p)$ over the observations in the leaf, the same quantity as in the gradient boosting leaf formula.

1. **In words:** add the residuals in the leaf and square the sum; divide by the sum of $p(1-p)$ over those observations plus λ .
2. **Formula:**

$$\text{similarity} = \frac{(\sum r_i)^2}{\sum p_i(1-p_i) + \lambda}$$

Here p_i is observation i 's predicted probability from the previous stage. As in regression, we take $\lambda = 0$.

3. **Example:** the root holds all five residuals; every $p_i = 0.6$, so $p_i(1-p_i) = 0.6 \times 0.4 = 0.24$:

$$\text{similarity}_{\text{root}} = \frac{(-0.6 + 0.4 - 0.6 + 0.4 + 0.4)^2}{5 \times 0.24} = \frac{0^2}{1.2} = 0$$

As with the mean in regression, the residuals from the base log-odds add up to exactly 0, so the root scores 0.

6. Choosing the split

Key point: Four midpoints, 5.975, 6.675, 7.625 and 8.875; the largest gain, 2.22, belongs to $\text{CGPA} < 7.625$.

The candidate thresholds are the midpoints of neighbouring CGPA values: $(5.70 + 6.25)/2 = 5.975$, then 6.675, 7.625 and 8.875. For each, the gain is the children's similarity minus the parent's, as in the [XGBoost regression Note](#), section 6.

Worked for $\text{CGPA} < 5.975$: student 1 alone on the left, the other four on the right.

$$S_{\text{left}} = \frac{(-0.6)^2}{0.24} = \frac{0.36}{0.24} = 1.5, \quad S_{\text{right}} = \frac{(0.4 - 0.6 + 0.4 + 0.4)^2}{4 \times 0.24} = \frac{0.36}{0.96} = 0.375$$

$$\text{gain} = 1.5 + 0.375 - 0 = 1.875$$

Split	Left residuals	Right residuals	S_{left}	S_{right}	Gain
CGPA < 5.975	-0.6	0.4, -0.6, 0.4, 0.4	1.50	0.38	1.88
CGPA < 6.675	-0.6, 0.4	-0.6, 0.4, 0.4	0.08	0.06	0.14
CGPA < 7.625	-0.6, 0.4, -0.6	0.4, 0.4	0.89	1.33	2.22
CGPA < 8.875	-0.6, 0.4, -0.6, 0.4	0.4	0.17	0.67	0.83

CGPA < 7.625 wins. Its right leaf holds only placed students; its left leaf holds both students who were not placed. We stop at depth 1: one split is enough to show the steps on five observations.

7. Output values in log-odds

Key point: A leaf's output is (sum of residuals) divided by $(\sum p(1-p) + \lambda)$: -1.11 for the left leaf and 1.67 for the right. These outputs are log-odds, so they can be added to the base log-odds.

1. **In words:** add the residuals in the leaf and divide by the sum of $p(1-p)$ over its observations plus λ .

2. **Formula:**

$$\text{output} = \frac{\sum r_i}{\sum p_i(1-p_i) + \lambda}$$

3. **Example:** the left leaf (students 1, 2, 3):

$$\text{output}_{\text{left}} = \frac{-0.6 + 0.4 - 0.6}{3 \times 0.24} = \frac{-0.8}{0.72} = -1.11$$

The right leaf (students 4, 5): $0.8/0.48 = 1.67$.

The output formula is the leaf formula of the [gradient boosting classification Note](#), now with λ added. Figure 1 shows the finished tree.

8. Stage 2: add in log-odds, read in probability

Key point: New log-odds = $0.405 + 0.3 \times \text{leaf output}$. The sigmoid turns it into a probability; the residuals shrink.

1. **In words:** add eta times the leaf output to the base log-odds, then apply the sigmoid.

2. **Formula:**

$$z^{(2)} = f_0 + \eta \cdot \text{tree}_1(x), \quad p^{(2)} = \frac{1}{1 + e^{-z^{(2)}}}$$

3. **Example:** student 1 (CGPA 5.70 < 7.625, left leaf), with $\eta = 0.3$:

$$z^{(2)} = 0.405 + 0.3 \times (-1.111) = 0.405 - 0.333 = 0.072, \quad p^{(2)} = \frac{1}{1 + e^{-0.072}} = 0.518$$

For students 4 and 5 (right leaf): $z^{(2)} = 0.405 + 0.3 \times 1.667 = 0.905$ and $p^{(2)} = 0.712$.

Student	Placed	Log-odds 2	Probability 2	Residual 2	Residual 1
1	0	0.072	0.518	-0.518	-0.6
2	1	0.072	0.518	0.482	0.4
3	0	0.072	0.518	-0.518	-0.6
4	1	0.905	0.712	0.288	0.4
5	1	0.905	0.712	0.288	0.4

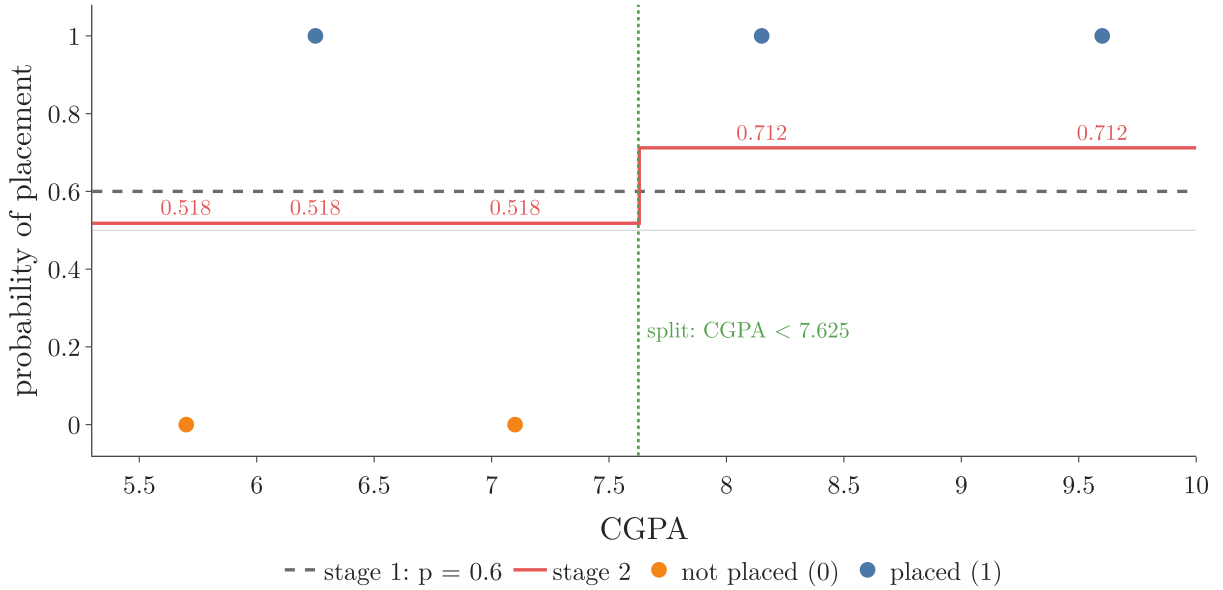


Figure 2: Stage 1 gives every student probability 0.6; after one tree the probability drops to 0.518 below CGPA 7.625 and rises to 0.712 above it

Four of the five residuals move towards 0 (Figure 2). Student 2 is placed but sits on the left with the two who were not, so its residual grows a little, from 0.4 to 0.482; later trees must fix that.

9. The next trees

Key point: Repeat with the new probabilities: residual 2 and $p^{(2)}(1 - p^{(2)})$ go into the next tree’s similarity scores and outputs.

Stage 3 grows a tree on CGPA and residual 2. The observations now have different probabilities, so the denominators change from one observation to the next: $0.518 \times 0.482 = 0.250$ on the left, $0.712 \times 0.288 = 0.205$ on the right. On our data the second tree prefers $\text{CGPA} < 5.975$ (gain 1.39), which isolates student 1.

The model after M trees is

$$z = f_0 + \eta \cdot \text{tree}_1(x) + \cdots + \eta \cdot \text{tree}_M(x), \quad p = \frac{1}{1 + e^{-z}}$$

and a new student is predicted “placed” when p is above the threshold, usually 0.5.

10. What changed from regression

Key point: Only the base model and the denominators. Everything else, splits, gains, eta, the loop, is the same.

	Regression	Classification
Base model	mean of the output	log-odds of class 1
Residual	$y - \hat{y}$	$y - p$
Similarity	$(\sum r)^2 / (n + \lambda)$	$(\sum r)^2 / (\sum p(1 - p) + \lambda)$
Output value	$\sum r / (n + \lambda)$	$\sum r / (\sum p(1 - p) + \lambda)$
Trees add up in	the output’s units (LPA)	log-odds

Extra: With $\lambda = 1$ (XGBoost’s default) the outputs shrink much more than in regression, because the denominators 0.72 and 0.48 are small next to 1: the leaves give $-0.8/1.72 = -0.47$ and $0.8/1.48 = 0.54$ instead of -1.11 and 1.67 .

Extra: XGBoost also requires every leaf to have $\sum p(1 - p)$ of at least `min_child_weight`, which is 1 by default. Here each observation contributes 0.24 and all five together only 1.2, so every possible split leaves one child below 1 (the best case is 0.48). XGBoost with default settings therefore does not split this toy data at all: the Notebook’s tree is a single leaf, and every probability stays 0.6. Since $p(1 - p)$ is at most 0.25 (at $p = 0.5$), every leaf needs at least 4 observations to reach 1, and more once the predictions move towards 0 or 1. To reproduce this Note, set `min_child_weight=0` (XGBoost docs, Parameters).

11. The same in code

Key point: The Notebook grows the tree with NumPy and checks it against the XGBoost library: same split, same gain, same stage-2 log-odds. scikit-learn's gradient boosting classifier agrees on this first tree.

Extra: `GradientBoostingClassifier(loss="log_loss", n_estimators=1, learning_rate=0.3, max_depth=1)` starts from the same log-odds, 0.405, and uses the same leaf outputs with $\lambda = 0$. Its tree picks splits by the squared error of the residuals rather than by XGBoost's gain. On this first tree every observation has the same $p(1-p)$, so both pick $\text{CGPA} < 7.625$ and give exactly the log-odds 0.072 and 0.905. From the second tree on the $p(1-p)$ values differ, so the two can choose different splits: XGBoost divides each leaf's $(\sum r)^2$ by $\sum p(1-p)$, the squared-error split by the number of observations. On our second tree they still agree (both split at $\text{CGPA} < 5.975$).

Python: The first tree in XGBoost, with every setting matched to this Note.

```
from xgboost import XGBClassifier

model = XGBClassifier(
    n_estimators=1,
    max_depth=1,
    learning_rate=0.3,          # eta
    reg_lambda=0,              # lambda
    gamma=0,
    min_child_weight=0,        # see the Extra above
    base_score=0.6,           # starting probability
    objective="binary:logistic",
    tree_method="exact",
)
model.fit(X, y)                # X: CGPA, y: 0 or 1
model.predict(X, output_margin=True) # log-odds
model.predict_proba(X)
model.get_booster().get_dump(with_stats=True)
```

For classification `base_score` is given as a probability; XGBoost converts it to the log-odds 0.405 itself. The log-odds come out as 0.072 and 0.905. The dump shows the split $f_0 < 7.625$ with `gain=2.22` and `cover=1.2`, the sum of $p(1-p)$ over the five observations. The leaves read -0.333 and 0.5 : the outputs -1.11 and 1.67 already multiplied by `eta`.

With `n_estimators=2` the library's second tree splits at $\text{CGPA} < 5.975$ with gain 1.39, as in section 9.

12. Summary

Step	What we compute	Our numbers
Base model	log-odds of class 1	$\ln 1.5 = 0.405$, $p = 0.6$
Residual	$y - p$	-0.6 or 0.4
Similarity	$(\sum r)^2 / (\sum p(1 - p) + \lambda)$	root 0
Best split	largest gain	CGPA < 7.625 , gain 2.22
Leaf outputs	$\sum r / (\sum p(1 - p) + \lambda)$	-1.11 and 1.67
Stage 2	$z = 0.405 + 0.3 \times \text{output}$, then sigmoid	$p = 0.518$ or 0.712

- XGBoost classification is gradient boosting classification with XGBoost's trees.
- Trees are fitted to probability residuals, but their outputs are log-odds.
- The similarity score and output use $\sum p(1 - p) + \lambda$ in the denominator.
- The sigmoid turns the summed log-odds into the final probability.

13. Sources

- Chen, T. and Guestrin, C. (2016). *XGBoost: A Scalable Tree Boosting System*. KDD 2016 (arXiv:1603.02754).
- XGBoost documentation, *XGBoost Parameters* (`min_child_weight`, `base_score`), xgboost.readthedocs.io.

14. Key terms

Term	Meaning
Similarity score (classification)	$(\text{sum of residuals})^2 / (\sum p(1 - p) + \lambda)$, with p the previous probabilities
Output value (classification)	sum of residuals / $(\sum p(1 - p) + \lambda)$, in log-odds
<code>min_child_weight</code>	Smallest allowed sum of $p(1 - p)$ (in regression: number of observations) in a leaf; default 1
<code>base_score</code>	XGBoost's starting prediction; a probability for classification
Cover	XGBoost's name for the sum of $p(1 - p)$ (in regression: the number of observations) in a node